Master of the Universe Unizar

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RECAP

- Radio allows us to study different phenomena
- Radio telescope resolution

  Theta = 1.22 * lambda / Effective_size

- Short baselines measure large angular scales.
- Long baselines measure compact structure.
- Visibilities ⇔ FT(Sky source brightness distribution)
- Phase is as important as amplitude.

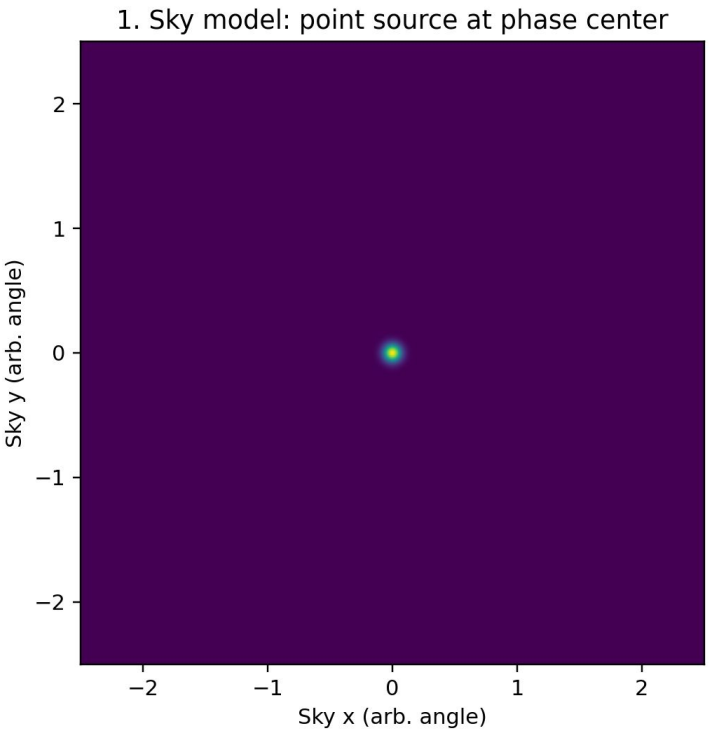
SUMMARY

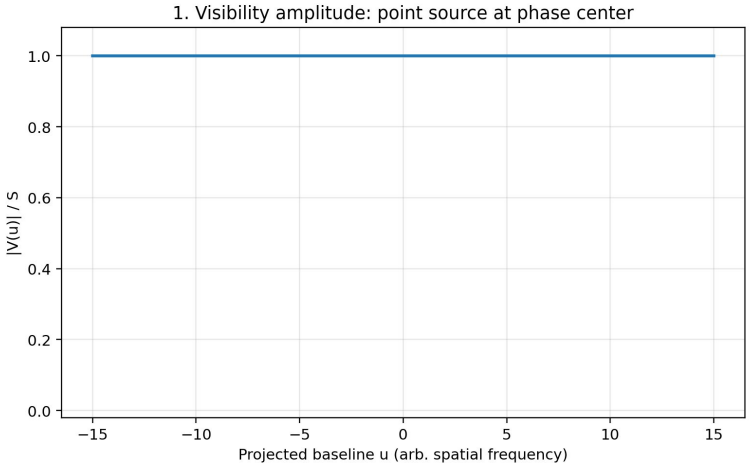
1. An interferometer does not directly make a normal image.
2. The first inverse FT gives a dirty image, not the true sky.
3. Dirty image = true sky convolved with the dirty beam.
4. CLEAN iteratively subtracts dirty-beam patterns from bright peaks.
5. Final image is a restored model, not a perfect direct photograph.

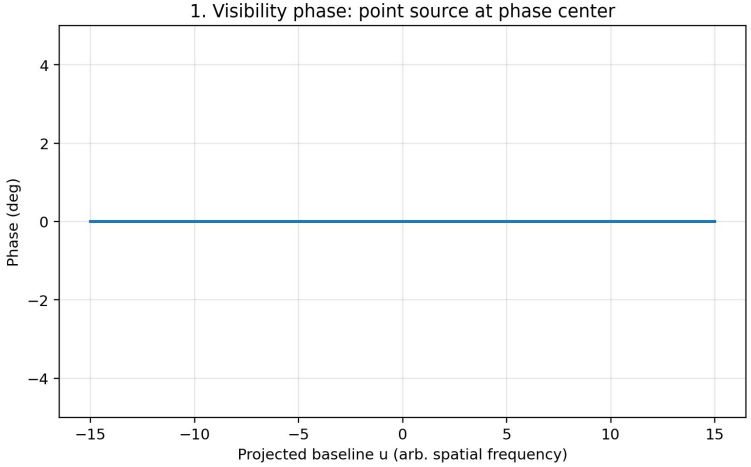
SKY MODELS and VISIBILITIES

Point source at phase centre

"Flat amplitude, zero phase."

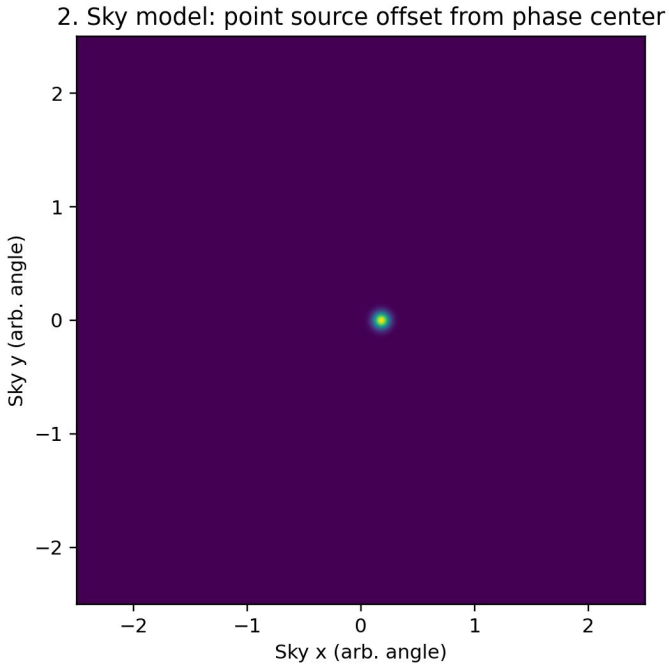


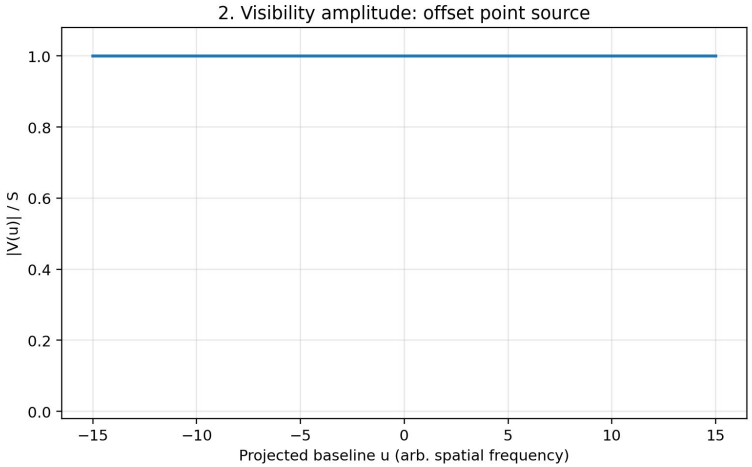


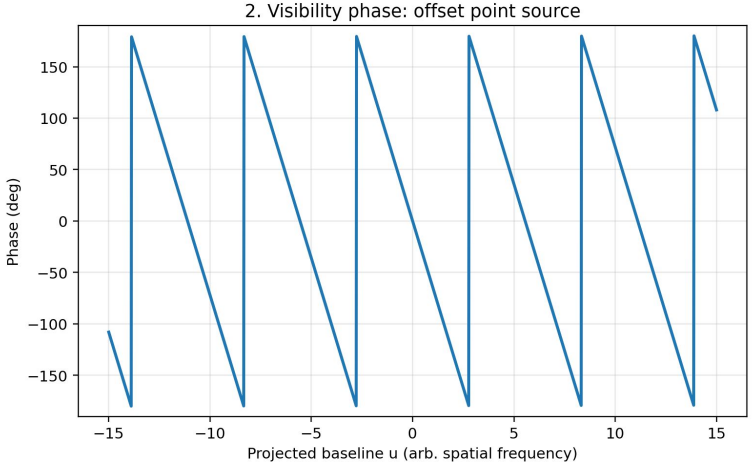


Offset point source

"Flat amplitude, phase ramp."

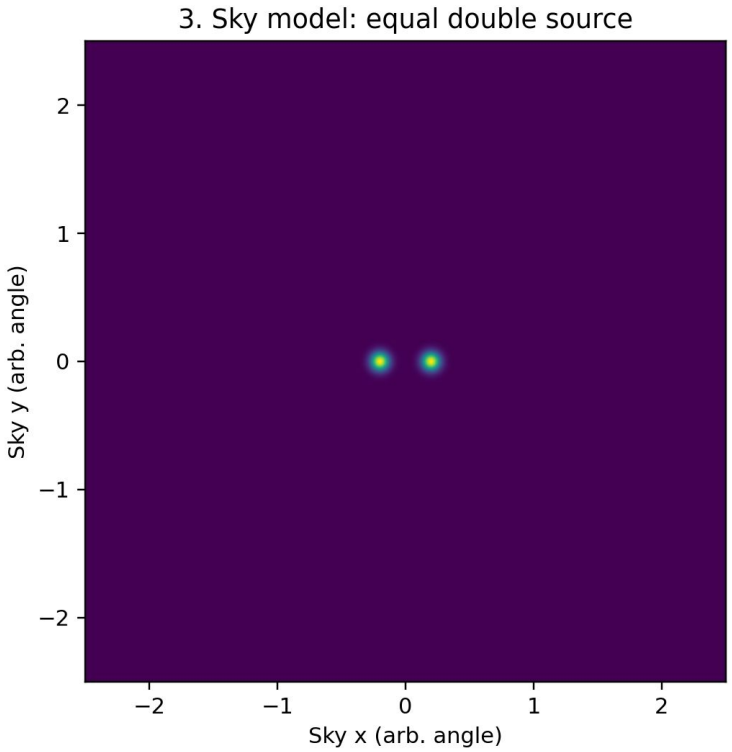


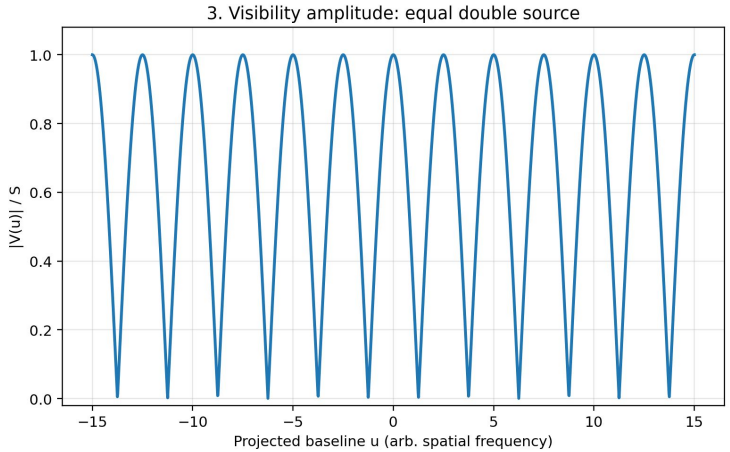


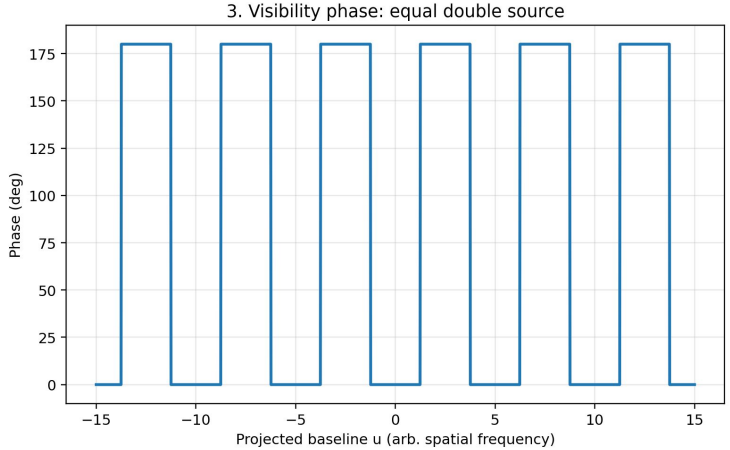


(Equal) double source

"Two components, oscillations and nulls."

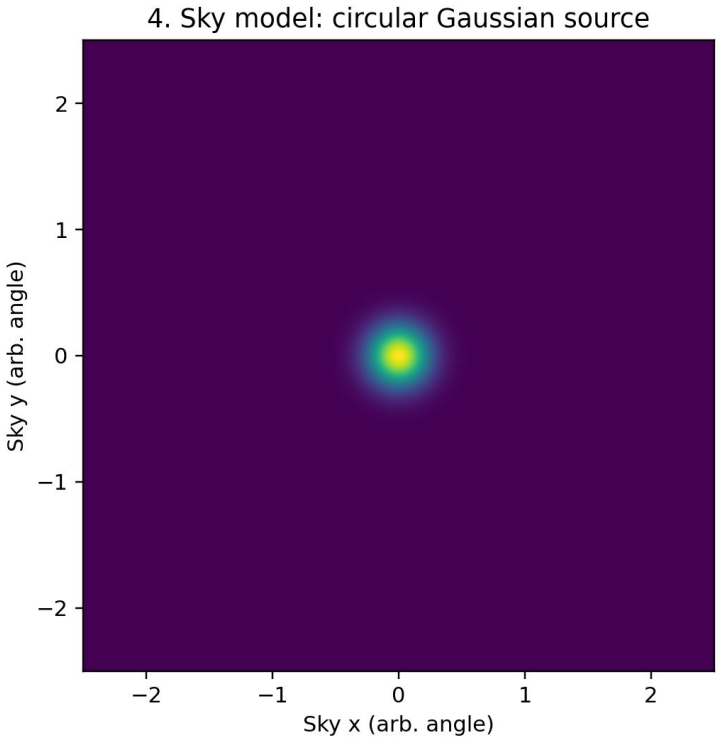


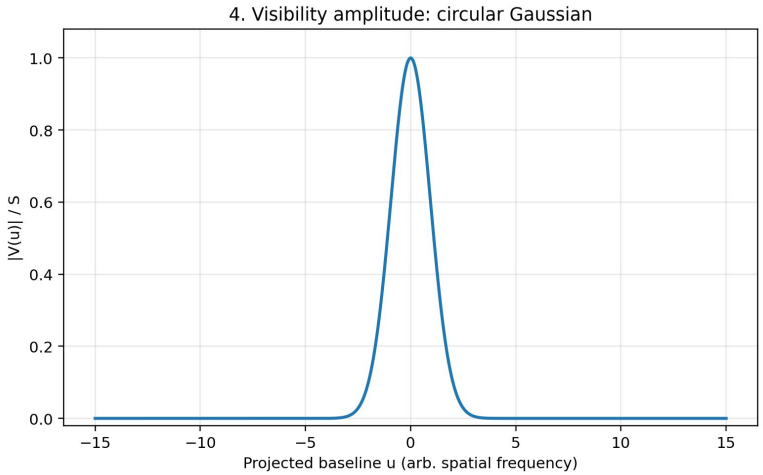


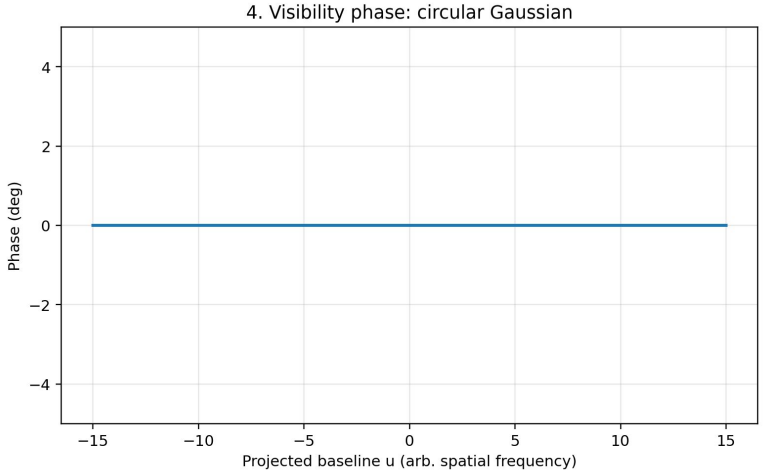


Gaussian

"Smooth source, smooth fall-off."

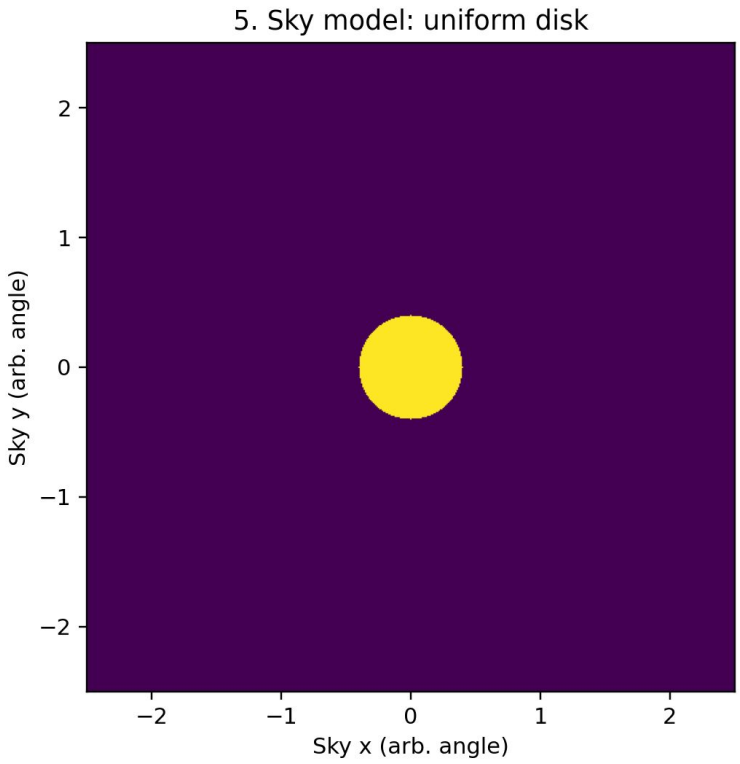


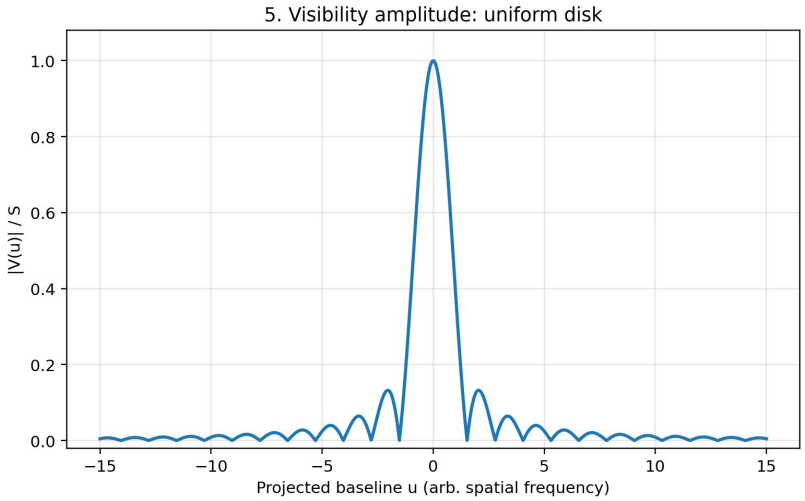


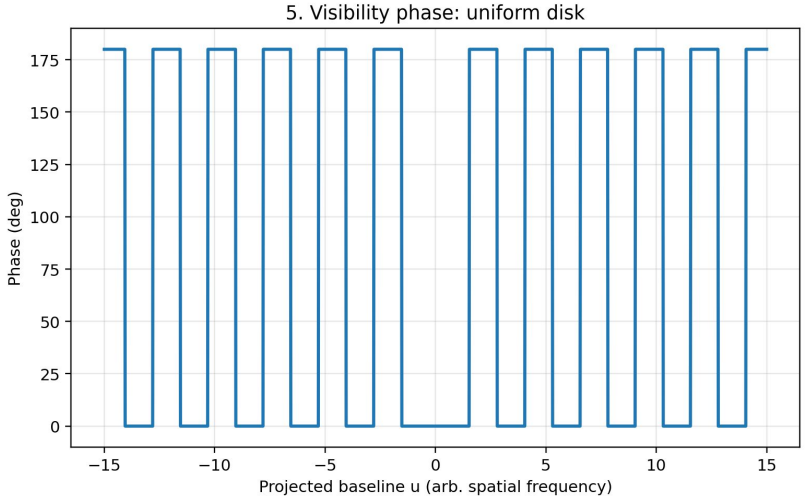


Uniform disk

"Sharp edge, Bessel-like nulls."

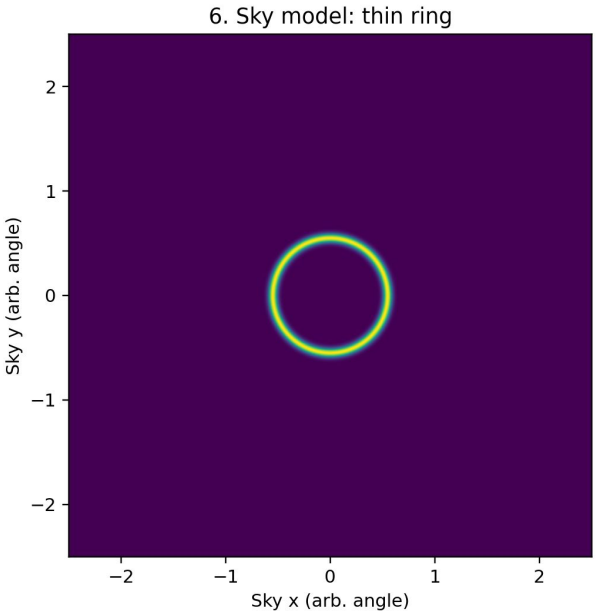


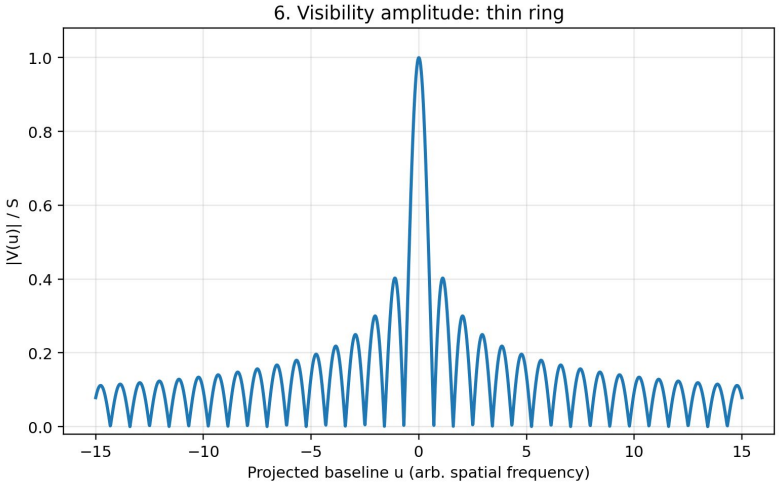


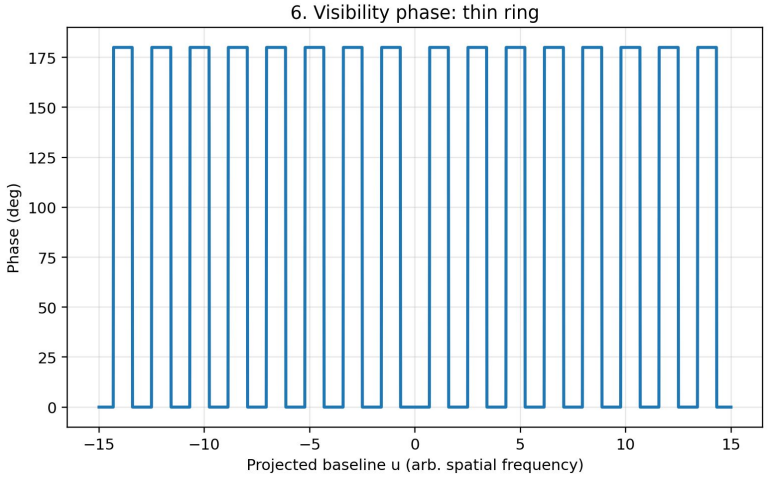


Thin ring

"More oscillatory, repeated nulls."

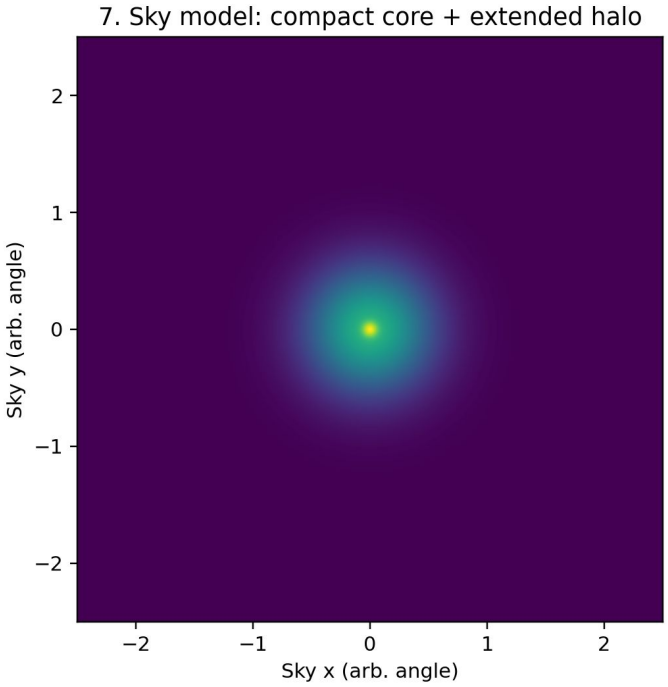


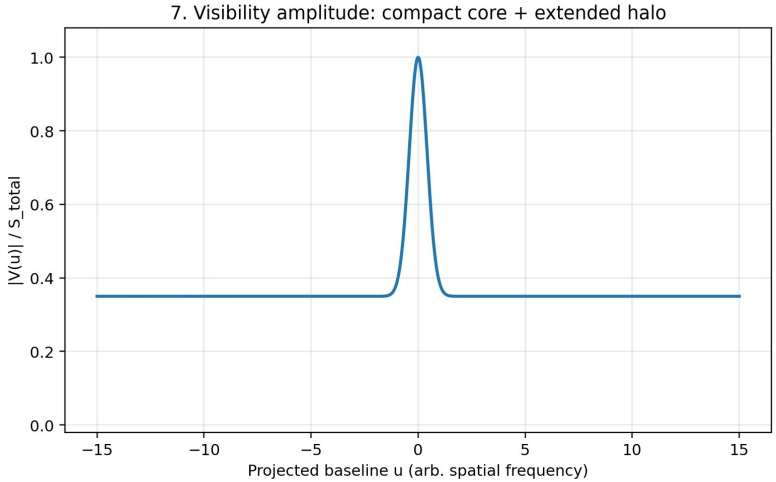


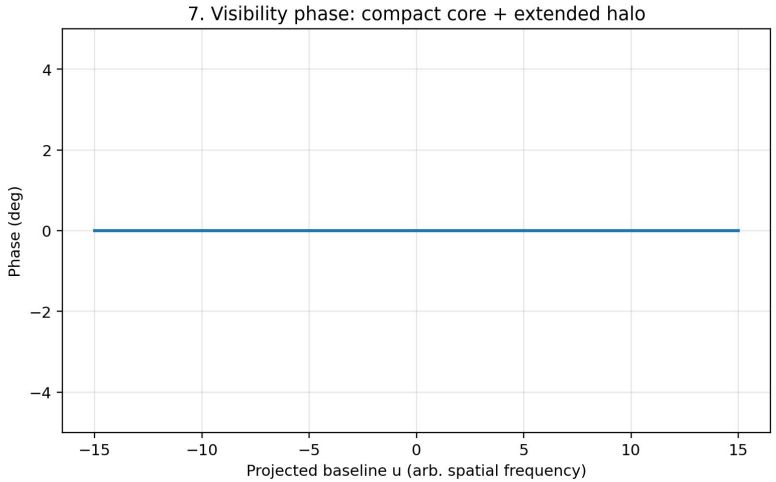


Core + halo

"Extended part resolved out; compact core survives."

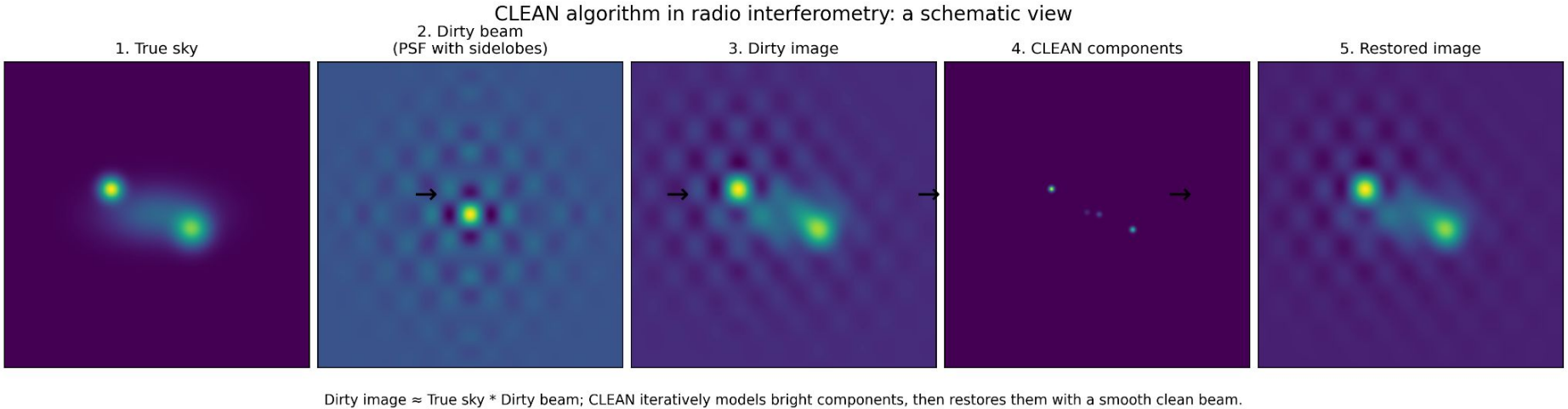






Radio interferometry imaging and the CLEAN algorithm

CLEAN in a nutshell



1. True sky brightness distribution.
2. Dirty beam = interferometer response to point source. Includes sidelobes due to incomplete sampling of u-v plane.
3. Dirty image: Inverse Fourier transform of measured visibilities. Real emission is there, but contaminated by dirty beam pattern (sidelobes)
4. CLEAN components: Bright peaks iteratively identified by the “CLEAN” algorithm.
5. “CLEAN-ed image”. Components convolved with idealized smooth clean beam + noise.

CLEAN in a nutshell

CLEAN models the sky as a set of components whose dirty-beam responses (=sidelobe contamination) are iteratively removed from the dirty image, and then it restores the model with a smooth beam plus residual noise.

- Figure is **schematic**, not a literal run of CLEAN on real visibilities
- In real observations, results depend on calibration quality, weighting, masking, stopping threshold, and source complexity